

Project Proposal: Mini Planet

Course: Introduction to Computer Graphics (2025/2026)

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1. Scenario

My project proposal consists in a mini earth like planet, with 3D terrain and "traces of civilization". The planet would be viewed from space.

2. Modeling and Animation

The application will demonstrate the modeling and animation of various objects, including the main planet and it's contents:

- **Objects:** Planet with 3d terrain, cities, houses, a cargo ship, a plane and a satellite.
- **Animations:** Automatic movements like the planet's rotation, satellite translation around the earth, plane flying around the earth, ship sailing.

3. Illumination and Textures

To enhance the visual composition of the project, I will add many lighting entities and materials to it's contents:

- **Illumination:** Stars and the sun in the background, the latter being an obvious light source and lights in the cities and houses that, when in the sun's shadow, will turn on.
- **Textures and Materials:** Not really sure if i should do something more realistic or cartoonish, so I don't really know if the materials used will be too complex, but they obviously must react to lighting.

4. Interactivity

The interactions I'll put in the project are the following:

- **Mouse:** Changing the user's perspective with a fixed position in the planet, rotating the planet around and clicking in various elements of the composition which will lead to animations or more interactivity.
- **Keyboard:** Controlling the ship and/or the plane.

Note: The things mentioned in this proposal are the most basic ideas for the project, some things might be changed or added later.